Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data columns tracked

37

Missing Values

In review ratings

Customer Data

- Age, Gender, Location
- Subscription Status
- Previous Purchases

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Discount Applied

Behavior Metrics

- Purchase Frequency
- Review Rating
- Shipping Type

Made with GAMMA

Data Preparation Process

1

Data Loading

Imported dataset using pandas

2

Initial Exploration

Checked structure and summary statistics

3

Missing Data

Imputed using median ratings by category

4

Standardization

Renamed columns to snake case

Feature Engineering

Age Groups

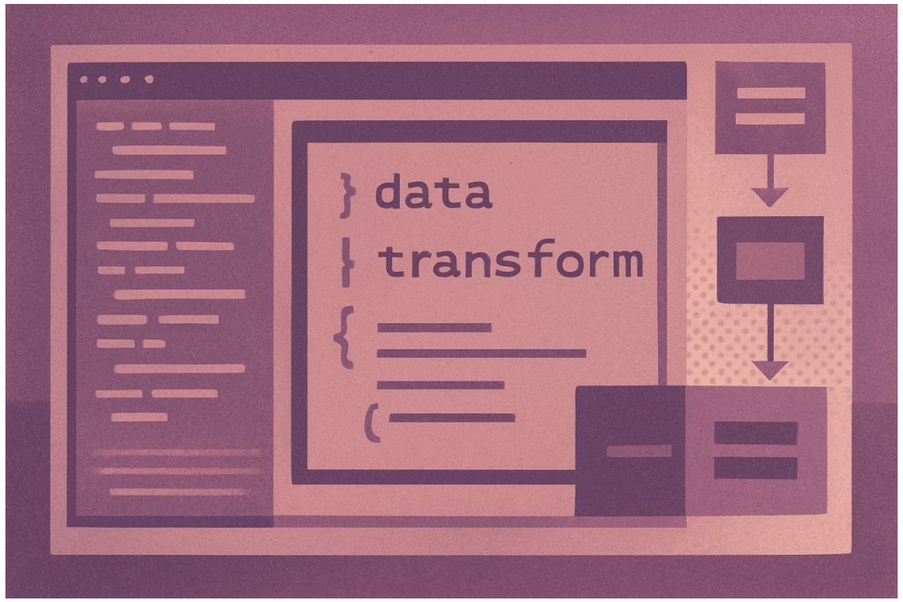
Created age_group column by binning customer ages

Purchase Frequency

Generated purchase_frequency_days from purchase data

Data Integration

Connected to PostgreSQL for SQL analysis



Made with GAMMA

SQL Analysis: Revenue Insights

Revenue by Gender

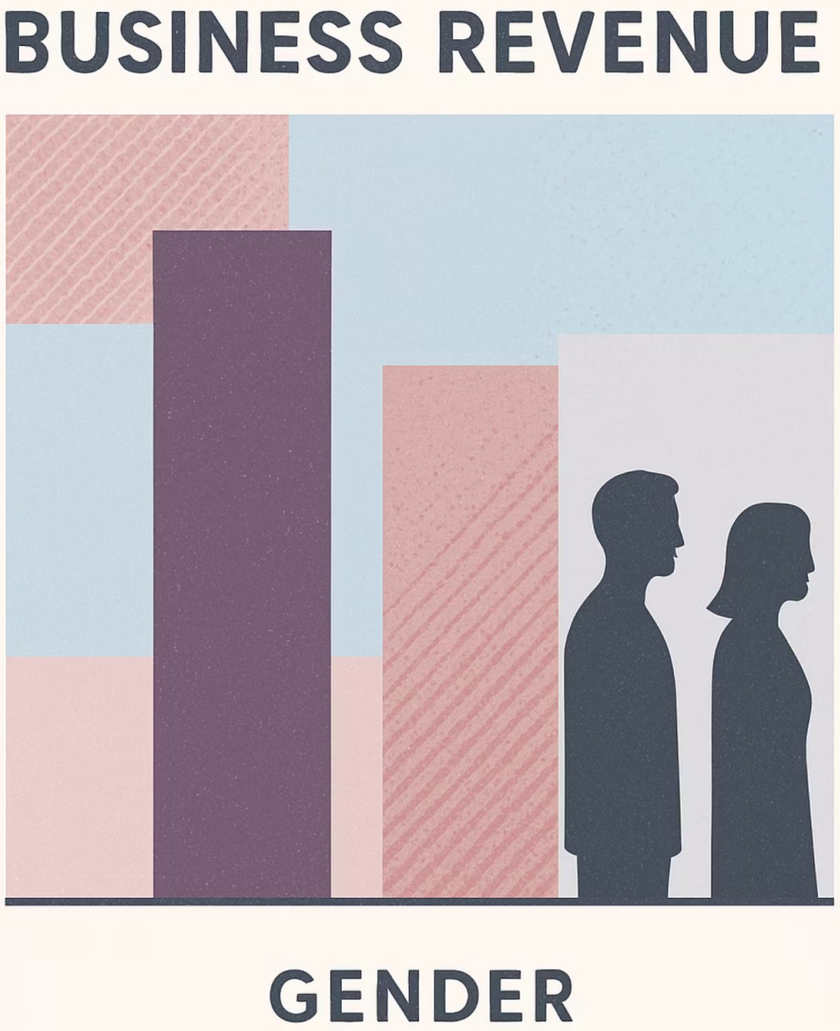
Compared total revenue generated by male vs. female customers

High-Spending Discount Users

Identified customers using discounts but spending above average

Top Products by Rating

Found products with highest average review ratings



SQL Analysis: Customer Behavior

01

Shipping Comparison

Compared average purchase amounts between Standard and Express shipping

02

Subscription Analysis

Compared average spend and total revenue across subscription status

03

Discount Patterns

Identified 5 products with highest percentage of discounted purchases

Made with GAMMA

SQL Analysis: Segmentation

Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history

Top Products per Category

Listed most purchased products within each category

Repeat Buyers

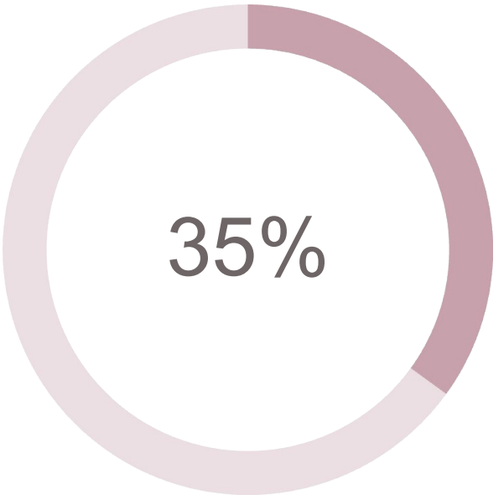
Checked if customers with >5 purchases are more likely to subscribe

Made with GAMMA

Power BI Dashboard

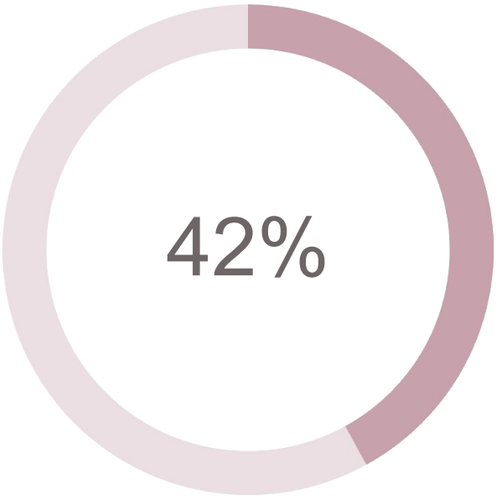
Interactive visualizations presenting key insights from customer shopping behavior analysis

Key Findings



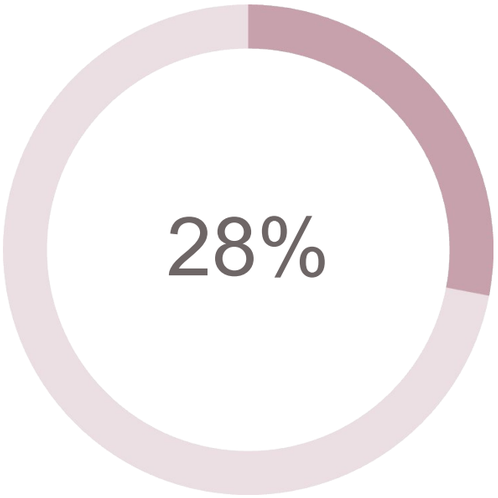
Discount Usage

Products with highest discount percentage



Express Shipping

Higher average purchase amounts



Subscription Rate

Among repeat buyers

Made with GAMMA

Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Made with GAMMA